

Signature and Cyclotomy in Real-Number Quantum Theory

*A split-Eisenstein companion to
Hoffreumon-Woods & Barrios Hita constructions*

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Abstract

Hoffreumon and Woods, together with the contemporaneous work of Barrios Hita and collaborators, recently established that quantum theory does not in fact require complex numbers. The Hoffreumon-Woods construction proceeds by twisting the tensor product so as to respect the algebra $J^2 = -I$ globally rather than locally, and privileges the Gaussian cyclotomic relation $x^2 + 1 = 0$, the discrete dihedral symmetry D_8 , and the positive-definite norm form $a^2 + b^2$ associated with the Brahmagupta-Fibonacci identity. The present note develops the structural question of what happens if one repeats their construction starting from a different cyclotomic relation. We work through the Eisenstein analogue ($x^2 + x + 1 = 0$, D_6 , positive-definite norm $a^2 - ab + b^2$) and the two split analogues, the standard split-complex ($x^2 - 1 = 0$, D_4 , indefinite norm $a^2 - b^2$) and a golden-ratio split-Eisenstein ($x^2 - x - 1 = 0$, infinite cyclic, indefinite norm $a^2 + ab - b^2$). The four constructions are organised along a signature axis with two end-points, Euclidean and Lorentzian, and the Hoffreumon-Woods construction sits at one extreme of this axis. We argue that the signature axis is the structurally meaningful generalisation of their construction, and we identify a signature tension in the chiral spinor formalism for gravity that the existence of split-cyclotomic real quantum theories puts in sharp relief. We close with a modular-arithmetic conjecture connecting the compact Hurwitz programme to a non-compact Lorentzian framework via the elliptic points of $SL(2, \mathbb{Z})$ on the upper half-plane.

1 Introduction

The mathematical question of whether quantum theory genuinely requires complex numbers has a long history, recently revived by Renou et al. [2] who argued that real-number quantum theory cannot reproduce all the predictions of the standard complex formulation. That argument was decisively answered in the negative by Hoffreumon and Woods [1], who exhibited an explicit real-number quantum theory consistent with the standard postulates and reproducing the same statistics as the complex theory. A closely related result was established in parallel by Barrios Hita, Trushechkin, Kampermann, Epping and Bruß [3], who introduced a physically motivated postulate about composite quantum systems implemented through local two-dimensional "flag" degrees of freedom carrying a redundant global phase,

with an equivalence relation on the multipartite flag state carrying the standard complex-phase ambiguity. The two constructions establish the consistency of real-number quantum theory by complementary routes: Hoffreumon and Woods through a categorical modification of the tensor-product representation, Barrios Hita et al. through a physically motivated postulate about composite system structure. The convergence of these two methodologically distinct routes on the same conclusion strengthens the structural claim that complex numbers in quantum mechanics are a matter of convenience rather than necessity. The construction of Hoffreumon and Woods [1] is built on a recognition that the Kronecker product is only one possible matrix representation of the tensor product, and that a suitably twisted alternative respects the algebra of complex numbers at the global rather than merely local level. The technical content of the twist is a new tensor-product representation, denoted $\otimes_r$, which embeds the algebra of complex matrices into the algebra of real matrices via the substitution $\{1, i\} \mapsto \{I, J\}$ where $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ satisfies $J^2 = -I$, and whose tensor structure ensures that this substitution behaves correctly under composition of subsystems.

The present paper takes as its starting point the structural observation that the Hoffreumon-Woods construction privileges one particular cyclotomic relation, namely $x^2 + 1 = 0$, out of a family of plausible candidates. This relation determines the matrix J that represents the imaginary unit, the discrete dihedral group D_8 generated by $\{I, J, X, Z\}$ that controls the algebra of the twisted tensor product, and the quadratic form $a^2 + b^2$ that gives the positive-definite Gaussian norm. Each of these three structural ingredients depends on the choice of $J^2 = -I$, and each would change if a different defining relation were chosen. The most immediate question, which we develop in detail below, is whether the same construction can be carried out for the Eisenstein cyclotomic relation $x^2 + x + 1 = 0$, the standard split-complex relation $x^2 - 1 = 0$, and the golden-ratio split-Eisenstein relation $x^2 - x - 1 = 0$. In each case the algebra works out cleanly at the scalar level; the four resulting candidate tensor products differ in their associated discrete symmetry groups and, decisively, in the signature of their norm forms.

The signature axis is, on the reading developed in this paper, the correct structural generalisation of the Hoffreumon-Woods construction. The two positive-definite cyclotomic flavours (Gaussian and Eisenstein) give compact discrete symmetries and finite cyclic unit groups; the two indefinite-signature flavours (split-complex and split-Eisenstein) give non-compact discrete symmetries and infinite unit groups. The former two correspond to the Euclidean wing of the duality identified in the companion paper [5] between volumetric and barycentric trace-free conditions; the latter two correspond to the Lorentzian wing. The signature tension this surfaces in the chiral spinor formalism for gravity is, we argue, structurally meaningful: the spinor sector is intrinsically Lorentzian, by virtue of $SL(2, \mathbb{C})$ being the double cover of the restricted Lorentz group, while the Hurwitz arithmetic that the asymmetric extension proposes for the trace summand is intrinsically Euclidean, by virtue of the Hurwitz quaternion order living inside Hamilton's quaternions with positive-definite norm. The split-cyclotomic constructions of this paper give a family of candidate signature-matched alternatives, at the cost of losing the finite-group structure that makes the Hurwitz unit group $2T$ compatible with the Standard Model fermion content.

The paper is organised as follows. Section 2 reviews the four cyclotomic flavours and their associated norm forms. Section 3 develops the four norm-composition identities, of which Brahmagupta-Fibonacci is the Gaussian instance. Section 4 constructs the twisted tensor products at the operator level, generalising the Hoffreumon-Woods $\otimes_r$. Section 5 examines the role of signature in self-adjointness and positivity, identifying the obstruction to direct extension of the Hoffreumon-Woods positivity proofs to the indefinite case. Section 6 discusses the signature tension in chiral gravity and the role of the modular group $SL(2, \mathbb{Z})$ as a candidate Lorentzian arithmetic framework. Section 7 collects conjectures.

2 Four cyclotomic flavours

We begin by recording the four cyclotomic relations we will work with, their associated rings of algebraic integers, unit groups, and norm forms. Each is a quadratic relation over $\mathbb{Z}$ defining a quadratic extension $\mathbb{Q}(\alpha)$ of the rationals, where α is a root of the relation. The four cases are distinguished by the sign of the discriminant of the quadratic polynomial, which controls whether the roots are real or complex, and the sign of the norm form, which controls whether it is positive-definite or indefinite.

2.1 The Gaussian flavour

The Gaussian cyclotomic relation is $x^2 + 1 = 0$, with roots $\pm i$ generating $\mathbb{Q}(i)$. The ring of algebraic integers is the Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, with norm form

$$N(a + bi) = a^2 + b^2, \quad (1)$$

positive-definite of signature $(2, 0)$. The unit group is $U(\mathbb{Z}[i]) = \{\pm 1, \pm i\} \cong \mathbb{Z}_4$, finite and compact. At the matrix level, the imaginary unit is represented by $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ with $J^2 = -I$, and the discrete dihedral group that the Hoffreumon-Woods construction respects is D_8 , the symmetry group of the square.

2.2 The Eisenstein flavour

The Eisenstein cyclotomic relation is $x^2 + x + 1 = 0$, with roots $\omega = e^{2\pi i/3}$ and $\omega^2 = e^{-2\pi i/3}$ generating $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. The ring of algebraic integers is the Eisenstein integers $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$, with norm form

$$N(a + b\omega) = a^2 - ab + b^2, \quad (2)$$

positive-definite of signature $(2, 0)$. The unit group is $U(\mathbb{Z}[\omega]) = \{\pm 1, \pm \omega, \pm \omega^2\} \cong \mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$, finite and compact. At the matrix level, ω is represented by the companion matrix

$$W = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad W^2 + W + I = 0, \quad W^3 = I, \quad (3)$$

the discrete dihedral group is $D_6 \cong S_3$, the symmetry group of the equilateral triangle.

2.3 The split-complex flavour

The split-complex (or hyperbolic, or perplex) cyclotomic relation is $x^2 - 1 = 0$, with roots ± 1 . As a ring this gives $\mathbb{Z}[\varepsilon]/(\varepsilon^2 - 1) \cong \mathbb{Z} \oplus \mathbb{Z}$, which is not a field; it has zero divisors, since $(1 + \varepsilon)(1 - \varepsilon) = 1 - \varepsilon^2 = 0$. The norm form is

$$N(a + b\varepsilon) = a^2 - b^2, \quad (4)$$

indefinite of signature $(1, 1)$. The "unit group" in the appropriate sense (elements of norm ± 1 that are not zero divisors) is the non-compact group $\{e^{\theta\varepsilon} : \theta \in \mathbb{R}\} \cup \{\pm 1, \pm \varepsilon\}$, which contains a $\mathbb{Z}_4$ subgroup of order four generated by elements like ε but is itself $\mathbb{R} \times \mathbb{Z}_2$. At the matrix level, ε is represented by the Pauli-style matrix

$$E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad E^2 = I, \quad (5)$$

and the relevant discrete symmetry generated by $\{I, E\}$ is the Klein four-group $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$, with an enlargement to D_4 when one adjoins a rotation.

2.4 The split-Eisenstein flavour

The split-Eisenstein cyclotomic relation is $x^2 - x - 1 = 0$, with roots $\varphi = \frac{1+\sqrt{5}}{2}$ (the golden ratio) and $\psi = \frac{1-\sqrt{5}}{2} = -1/\varphi$. Both roots are real, and $\mathbb{Q}(\varphi) = \mathbb{Q}(\sqrt{5})$ is a real quadratic field. The ring of algebraic integers is $\mathbb{Z}[\varphi] = \{a + b\varphi : a, b \in \mathbb{Z}\}$, with norm form computed via the Galois conjugation $\sigma : \varphi \leftrightarrow \psi$:

$$N(a + b\varphi) = (a + b\varphi)(a + b\psi) = a^2 + ab(\varphi + \psi) + b^2\varphi\psi = a^2 + ab - b^2, \quad (6)$$

where we used $\varphi + \psi = 1$ and $\varphi\psi = -1$. The discriminant of this form is $1 + 4 = 5 > 0$, so it is indefinite of signature $(1, 1)$. The unit group $U(\mathbb{Z}[\varphi]) = \{\pm\varphi^n : n \in \mathbb{Z}\} \cong \mathbb{Z} \times \mathbb{Z}_2$ is infinite and non-compact, with φ generating an infinite cyclic subgroup. At the matrix level, φ is represented by the companion matrix

$$\Phi = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Phi^2 - \Phi - I = 0, \quad (7)$$

and the discrete symmetry generated by $\{I, \Phi\}$ together with the Galois reflection is the infinite dihedral group D_∞ .

2.5 The signature axis

The four cyclotomic flavours line up along a single structural axis, which we record as Table 1.

Flavour	Relation	Roots	Norm form	Signature	Unit group
Gaussian	$x^2 + 1 = 0$	$\pm i$	$a^2 + b^2$	$(2, 0)$	$\mathbb{Z}_4$
Eisenstein	$x^2 + x + 1 = 0$	ω, ω^2	$a^2 - ab + b^2$	$(2, 0)$	$\mathbb{Z}_6$
Split-complex	$x^2 - 1 = 0$	± 1	$a^2 - b^2$	$(1, 1)$	$\mathbb{R} \times \mathbb{Z}_2$
Split-Eisenstein	$x^2 - x - 1 = 0$	φ, ψ	$a^2 + ab - b^2$	$(1, 1)$	$\mathbb{Z} \times \mathbb{Z}_2$

Table 1: The four cyclotomic flavours, arranged by signature. Positive-definite flavours (top two rows) have compact finite unit groups; indefinite-signature flavours (bottom two rows) have non-compact infinite unit groups. The Eisenstein-vs-Gaussian distinction within each signature is a barycentric-vs-volumetric one; see Section 6.

The structural pattern visible in the table is the following. Within each signature class, there is a Gaussian-style flavour with a $\mathbb{Z}_4$ -related symmetry and an Eisenstein-style flavour with a $\mathbb{Z}_3$ -related symmetry. The signature axis distinguishes compact positive-definite arithmetic from non-compact indefinite arithmetic. The Hoffreumon-Woods construction takes the Gaussian flavour as input; the natural Eisenstein companion construction takes the Eisenstein flavour; and the split versions take the indefinite flavours. The four twisted tensor products that result are organised by signature in a way that we make precise in Section 4.

3 Norm composition identities

Each cyclotomic flavour comes with an associated norm-composition identity, expressing the multiplicativity of the norm form on products. These identities are the scalar-level content of the twisted tensor products that we will construct at the operator level in Section 4.

3.1 Brahmagupta-Fibonacci

The Gaussian norm satisfies the classical Brahmagupta-Fibonacci identity:

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2, \quad (8)$$

corresponding to the product $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$ in $\mathbb{Z}[i]$. This identity has been known in some form since at least Diophantus, was systematised by Brahmagupta in the seventh century in the more general form for indefinite Pell-like quadratic forms, and was rediscovered in connection with Fibonacci's *Liber Quadratorum* in the thirteenth century [6]. Its modern statement, that the norm on $\mathbb{Z}[i]$ is a multiplicative quadratic form, is precisely the condition that the substitution $\{1, i\} \mapsto \{I, J\}$ extends to a ring homomorphism into the algebra of real matrices.

3.2 The Eisenstein composition

The Eisenstein norm satisfies the analogous identity:

$$(a^2 - ab + b^2)(c^2 - cd + d^2) = (ac - bd)^2 - (ac - bd)(ad + bc - bd) + (ad + bc - bd)^2, \quad (9)$$

corresponding to the product $(a + b\omega)(c + d\omega) = (ac - bd) + (ad + bc - bd)\omega$ in $\mathbb{Z}[\omega]$, where we use $\omega^2 = -1 - \omega$ to eliminate ω^2 terms. This identity expresses the multiplicativity of the Eisenstein norm and is the Eisenstein analogue of Brahmagupta-Fibonacci.

3.3 The split-complex composition

The split-complex norm satisfies a corresponding indefinite version of Brahmagupta-Fibonacci:

$$(a^2 - b^2)(c^2 - d^2) = (ac + bd)^2 - (ad + bc)^2, \quad (10)$$

corresponding to the product $(a + b\varepsilon)(c + d\varepsilon) = (ac + bd) + (ad + bc)\varepsilon$ with $\varepsilon^2 = 1$. The structural difference from (8) is the sign flip on the cross-term bd in the first coordinate (positive rather than negative), reflecting $\varepsilon^2 = +1$ rather than $i^2 = -1$.

3.4 The split-Eisenstein composition

The golden-ratio norm form satisfies the indefinite analogue of (9):

$$(a^2 + ab - b^2)(c^2 + cd - d^2) = (ac + bd)^2 + (ac + bd)(ad + bc + bd) - (ad + bc + bd)^2, \quad (11)$$

corresponding to the product $(a + b\varphi)(c + d\varphi) = (ac + bd) + (ad + bc + bd)\varphi$ in $\mathbb{Z}[\varphi]$, where we used $\varphi^2 = \varphi + 1$ to eliminate φ^2 terms. This is the genuinely new identity in our setting. It expresses the multiplicativity of the golden-ratio norm form, and it is the scalar-level content of the split-Eisenstein twisted tensor product to be constructed in Section 4.

Observation 3.1. The four composition identities (8)–(11) are formally identical up to sign changes that track the cyclotomic relation. Each takes the form

$$[Q(a, b)][Q(c, d)] = Q(P_1(a, b, c, d), P_2(a, b, c, d)),$$

where Q is the relevant quadratic form and P_1, P_2 are bilinear polynomials. The signs of the coefficients in Q and in P_1, P_2 are determined by the cyclotomic relation $\alpha^2 = c_1\alpha + c_0$, with $c_0, c_1 \in \mathbb{Z}$ giving the coefficients of the norm form $Q(a, b) = a^2 - c_1ab - c_0b^2$. The cases collected here correspond to $(c_1, c_0) \in \{(0, -1), (-1, -1), (0, 1), (1, 1)\}$ for Gaussian, Eisenstein, split-complex, and split-Eisenstein respectively.

4 Twisted tensor products at the operator level

The Hoeffreumon-Woods construction lifts the Brahmagupta-Fibonacci identity (8) from the scalar level to the operator level, by defining a tensor product $\otimes_r$ on the space of "special symmetric" real matrices, which is the image of the Hermitian complex matrices under the substitution

$$\Gamma : \text{Herm}_n(\mathbb{C}) \rightarrow \text{Sym}_{2n}(\mathbb{R}), \quad H^{\text{Sy}} + iH^{\text{An}} \mapsto I \otimes_K H^{\text{Sy}} + J \otimes_K H^{\text{An}}, \quad (12)$$

with $\otimes_K$ the ordinary Kronecker product. We refer the reader to [1] for the detailed construction, and for the proof that the resulting tensor product yields a real-number quantum theory consistent with the standard postulates and reproducing the statistics of the complex theory.

The structural feature we extract from their construction is that the recipe is parametric in the choice of cyclotomic generator. For each of the four flavours of Section 2, the same recipe yields a candidate twisted tensor product, distinguished by the matrix representation of the cyclotomic generator and by the discrete symmetry group that the resulting tensor product respects.

4.1 The Gaussian twisted tensor product

This is the Hoeffreumon-Woods $\otimes_r$, reviewed for completeness. The matrix J of (12) represents the imaginary unit, and the twisted tensor product is defined so that the relation $J \otimes_r J = -I \otimes_r I$ holds globally at the operator level, mirroring the scalar relation $i \cdot i = -1$. The discrete symmetry that the construction respects is D_8 , generated by $\{I, J, X, Z\}$ with $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

4.2 The Eisenstein twisted tensor product

The Eisenstein analogue of $\otimes_r$ replaces J by the cube-root matrix W of (3). The cyclotomic relation respected globally is $W^2 + W + I = 0$, and the discrete symmetry group is $D_6 \cong S_3$, generated by $\{I, W, R\}$ where R is a reflection satisfying $R^2 = I$ and $RWR = W^2$. An explicit choice is $R = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}$, which one verifies satisfies $R^2 = I$ and $RWR = W^2$.

The recipe for the Eisenstein twisted tensor product follows by analogy with the Hoeffreumon-Woods construction, replacing the Hermitian splitting $H = H^{\text{Sy}} + iH^{\text{An}}$ with a $\mathbb{Z}[\omega]$ -module splitting $H = H_0 + WH_1$, where H_0, H_1 are real matrices of compatible dimension. The detailed verification of positivity, locality, and statistical agreement (the analogues of the Hoeffreumon-Woods main theorems) is a separate technical undertaking which we do not carry out here; we record only the structural shape of the construction.

4.3 The split-complex twisted tensor product

The split-complex analogue of $\otimes_r$ replaces J by the square-root-of-unity matrix E of (5). The cyclotomic relation respected globally is $E^2 = I$, and the discrete symmetry group is the Klein four-group $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$, with a dihedral enlargement to D_4 when one includes a rotation J that anticommutes with E .

The structural feature distinguishing this construction from the Gaussian and Eisenstein cases is that the norm form is indefinite. The "Hermitian-like" splitting $H = H^{\text{Sy}} + \varepsilon H^{\text{An}}$ no longer produces a positive-definite cone of operators in the usual way, because the squared norm $\|H^{\text{Sy}} + \varepsilon H^{\text{An}}\|^2 = \|H^{\text{Sy}}\|^2 - \|H^{\text{An}}\|^2$ can be negative. The notion of "state" in the resulting twisted tensor product space requires reformulation, and we return to this in Section 5.

4.4 The split-Eisenstein twisted tensor product

The golden-ratio analogue of $\otimes_r$ replaces J by the matrix Φ of (7). The cyclotomic relation respected globally is $\Phi^2 - \Phi - I = 0$, and the discrete symmetry group is the infinite dihedral group D_∞ , generated by $\{I, \Phi, R\}$ where R implements the Galois conjugation $\varphi \leftrightarrow \psi$ exchanging the golden ratio with its conjugate.

This construction is, like the split-complex case, signature-mismatched relative to the Hoffreumon-Woods construction. The norm form is indefinite, the unit group is non-compact, and the standard Hermitian-style positivity arguments do not apply. The structural appeal of the construction is that it is the Eisenstein analogue of the split-complex construction, just as the standard Eisenstein construction is the Eisenstein analogue of the Hoffreumon-Woods construction. The four constructions therefore organise into a 2×2 grid by signature and cyclotomic generation, with Hoffreumon-Woods at one corner.

	Volumetric ($\mathbb{Z}_4$ -style)	Barycentric ($\mathbb{Z}_3$ -style)
Definite signature	Hoffreumon-Woods $J^2 = -I, D_8$	Eisenstein companion $W^2 + W + I = 0, D_6$
Indefinite signature	Split-complex $E^2 = I, D_4$	Split-Eisenstein $\Phi^2 = \Phi + I, D_\infty$

5 Self-adjointness, positivity, and signature

The Hoffreumon-Woods construction succeeds in reproducing the standard quantum statistics in part because the splitting $H = H^{\text{Sy}} + iH^{\text{An}}$ of a complex Hermitian matrix into real symmetric and real antisymmetric parts has a clean lift to the real-matrix setting under the substitution $i \mapsto J$. The resulting "special symmetric" subspace inherits a natural positive cone, and the inner product structure on this cone reproduces the Hilbert-Schmidt inner product of the original complex setting. Positivity is preserved under the twisted tensor product, and the resulting real-number quantum theory therefore has states, effects, and Born-rule probabilities that match those of the complex theory exactly.

The structural feature that makes this work is the positive-definiteness of the Gaussian norm form. The two pieces H^{Sy} and H^{An} both contribute positively to the squared Hilbert-Schmidt norm, with $\|H\|^2 = \|H^{\text{Sy}}\|^2 + \|H^{\text{An}}\|^2$, and the positivity of $\Gamma(H)$ as a real symmetric matrix is therefore equivalent to the positivity of H as a complex Hermitian matrix. The clean correspondence between these two notions of positivity is what allows the Hoffreumon-Woods construction to reproduce the standard statistics.

In the Eisenstein case, the same structural feature holds. The Eisenstein norm form $a^2 - ab + b^2$ is positive-definite, with discriminant $1 - 4 = -3 < 0$. The splitting of an Eisenstein-Hermitian matrix into the real and Eisenstein parts under $\omega \mapsto W$ admits a clean positivity correspondence, although the analogue of the Hoffreumon-Woods Hermitian condition has to be formulated carefully because the natural Galois involution on $\mathbb{Z}[\omega]$ sends ω to $\omega^2 = \bar{\omega}$ rather than to $-\omega$. The appropriate notion of "Eisenstein self-adjointness" replaces Hermitian invariance with cyclic-invariance under the $\mathbb{Z}_3$ generator W (or with σ -invariance under the Galois conjugation), and the resulting cone of positive elements is structurally analogous to the Hoffreumon-Woods case. We expect, although we do not prove here, that the Eisenstein construction succeeds in reproducing the standard quantum

statistics in the same way that the Hoffreumon-Woods construction does. Whether this expectation is correct is a question for separate work.

5.1 The breakdown in indefinite signatures

For the two indefinite-signature constructions (split-complex and split-Eisenstein), the positivity structure fails in a structurally significant way. The squared norm $\|H^{Sy}\|^2 - \|H^{An}\|^2$ of the split-complex case is indefinite, with positive contributions from one sector and negative contributions from the other. There is no automatic correspondence between the positivity of a matrix in the split-complex sense and the positivity of its image under the analogue of Γ . The positive cone in the split-complex setting is not the cone of positive semi-definite real symmetric matrices, but a quite different cone defined by an indefinite inner product.

The same structural feature recurs in the split-Eisenstein case, with the golden-ratio norm form $a^2 + ab - b^2$ replacing the split-complex $a^2 - b^2$. The cone of "states" in either of these settings would have to be defined by appeal to the indefinite inner product rather than the Frobenius inner product, and the resulting notion of probability would be substantially different from the Born rule of the standard theory.

5.2 Krein-space reformulation

A natural mathematical framework for indefinite-signature operator theories is the Krein space [7], which is a Hilbert space equipped with an indefinite inner product and a distinguished fundamental decomposition into positive and negative sectors. Operators on a Krein space can be Hermitian-with-respect-to the indefinite inner product, and there is a developed spectral theory for such operators. A split-cyclotomic real quantum theory built using $\otimes_r$ -style twisted tensor products on Krein spaces would naturally inherit a Krein-Hermitian structure rather than an ordinary Hermitian one.

Whether such a construction can be made to reproduce a recognisable form of quantum statistics is a question that, to our knowledge, has not been explored in the literature. The closest analogue we are aware of is the use of Krein spaces in pseudo-Hermitian quantum mechanics [8,9], where the underlying Hilbert space is complex but the operators of interest are Krein-Hermitian rather than ordinary Hermitian. A split-cyclotomic real quantum theory would generalise this in the direction of using a real-matrix substrate rather than a complex one.

Question 5.1. Does there exist a notion of "state" on a split-cyclotomic real quantum theory, defined via an appropriate Krein-positivity cone, such that the twisted tensor product preserves it and the resulting probabilistic structure reproduces a recognisable physical theory?

We do not attempt to answer this question here. Our purpose is to identify it as the structurally meaningful question that the existence of split-cyclotomic constructions raises.

6 Signature tension in chiral gravity and the modular bridge

The signature axis of Section 2 acquires a physical significance when one considers the chiral first-order formalism for gravity in the Plebański-Penrose-Rindler tradition. The standard formulation takes the gauge group to be $GL(2, \mathbb{C})$ and decomposes the gauge algebra as

$$\mathfrak{gl}(2, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathbb{C}, \quad (13)$$

with the $\mathfrak{sl}(2, \mathbb{C})$ summand carrying the gravitational degrees of freedom and the abelian $\mathbb{C}$ summand carrying the density-weight data. The asymmetric Hurwitz extension of [5]

proposes to replace the abelian summand $\mathbb{C}$ by the Hurwitz quaternion order $\mathbb{H}_H$, while leaving $\mathfrak{sl}(2, \mathbb{C})$ intact, on the grounds that the resulting extended algebra carries both a volumetric trace-free condition on its $\mathfrak{sl}(2, \mathbb{C})$ sector and a barycentric trace-free condition on its arithmetic sector, with the latter brought in by the embedded Eisenstein $\mathbb{Z}_3 \subset \mathbb{H}_H^\times$.

6.1 The mismatch

The signature character of $\mathfrak{sl}(2, \mathbb{C})$ is intrinsically Lorentzian: $\mathrm{SL}(2, \mathbb{C})$ is the double cover of $\mathrm{SO}^+(3, 1)$, and the determinant condition $\det = 1$ corresponds at the Lie-algebra level to the trace-free condition $\mathrm{tr} = 0$ acting on $\mathbb{C}^2$ with its inherited indefinite-signature structure when the spacetime is Lorentzian. The Hurwitz quaternion order $\mathbb{H}_H$, by contrast, is intrinsically Euclidean: it lives inside Hamilton's quaternions $\mathbb{H} \cong \mathbb{R}^4$ with positive-definite norm $a^2 + b^2 + c^2 + d^2$, its unit group is the finite binary tetrahedral group $2T$ of order 24, and its vertex configuration is the regular 24-cell.

The two pieces of the extended algebra $\mathfrak{sl}(2, \mathbb{C}) \oplus \mathbb{H}_H$ therefore live on ambient spaces with different signature characters. The spinor sector is complex-Lorentzian; the arithmetic sector is Euclidean. The volumetric and barycentric constraints identified in the companion paper as the two trace-free conditions carried by the extended algebra act on ambient spaces with different metrics. Whether this signature mismatch is a feature or a bug of the construction is an open structural question.

6.2 The split-Eisenstein alternative

The split-cyclotomic constructions of Section 4 offer a signature-matched alternative. A "hyperbolic Hurwitz" programme based on the split quaternions $M(2, \mathbb{R})$ rather than Hamilton's $\mathbb{H}$ would naturally inherit the indefinite signature of the spinor sector, with the maximal order being $M(2, \mathbb{Z})$ and the unit group being $\mathrm{GL}(2, \mathbb{Z})$. Within this framework, the split-Eisenstein flavour developed above would replace the standard Eisenstein $\mathbb{Z}[\omega]$ by the golden-ratio $\mathbb{Z}[\varphi]$, with the indefinite norm $a^2 + ab - b^2$ inherited from the ambient indefinite signature.

The cost of this signature-matching is the loss of the finite-group structure that makes $2T$ compatible with the Standard Model fermion content. The unit group of $M(2, \mathbb{Z})$ is $\mathrm{GL}(2, \mathbb{Z})$, which is infinite and contains the modular group $\mathrm{SL}(2, \mathbb{Z})$ as its determinant-one subgroup. There is no finite analogue of the seven irreducible representations of $2T$ with dimensions $(1, 1, 1, 2, 2, 2, 3)$ saturating $\sum d_i^2 = 24$, and the Standard Model identification developed in the asymmetric Hurwitz paper has no obvious counterpart in the split setting.

6.3 The modular bridge

The two programmes are not, however, structurally unrelated. The modular group $\mathrm{SL}(2, \mathbb{Z})$ acts on the upper half-plane $\mathbb{H} = \{\tau \in \mathbb{C} : \mathrm{Im}(\tau) > 0\}$, and the quotient $\mathbb{H}/\mathrm{SL}(2, \mathbb{Z})$ is the moduli space of elliptic curves over $\mathbb{C}$. This space carries two distinguished elliptic points where the action has finite stabilisers:

- $\tau = i$, with stabiliser $\mathbb{Z}_4$ generated by the modular transformation $\tau \mapsto -1/\tau$. This is the Gaussian point.
- $\tau = \omega = e^{2\pi i/3}$, with stabiliser $\mathbb{Z}_3$ generated by $\tau \mapsto -1/(\tau + 1)$. This is the Eisenstein point.

The two compact-arithmetic flavours of Section 2 are therefore visible inside the non-compact modular framework as the local stabilisers at the two elliptic points of the modular curve.

The compactness of $\mathbb{Z}_4$ and $\mathbb{Z}_3$ is a feature of these specific elliptic points; the global action of $\mathrm{SL}(2, \mathbb{Z})$ on $\mathbb{H}$ is non-compact and hyperbolic.

This suggests the structural picture in which the Hurwitz programme is naturally seen as a local feature of a larger Lorentzian modular framework. The finite Hurwitz unit group $2T$ would appear as the local stabiliser-like structure at an Eisenstein-style point, embedded inside a non-compact modular ambient that carries the Lorentzian signature of the spinor sector. The signature mismatch identified in Section 6 would then be resolved by recognising that the Euclidean Hurwitz refinement is a local feature of an ambient Lorentzian theory, rather than a global signature property of the full extended algebra.

Question 6.1. Can the asymmetric Hurwitz extension of $\mathfrak{gl}(2, \mathbb{C})$ be embedded in a larger Lorentzian-modular framework in which the Hurwitz arithmetic appears as a local elliptic-point structure rather than as a global signature commitment? Concretely: is there a real-number quantum theory based on $\mathrm{SL}(2, \mathbb{Z})$ -twisted tensor products that contains the Hoffreumon-Woods construction as a local feature at $\tau = i$ and the Eisenstein companion construction as a local feature at $\tau = \omega$?

The resolution to this question would settle the signature-tension question in the asymmetric Hurwitz programme one way or the other.

7 Open problems and conjectures

We collect here the structural questions raised in the body of the paper, together with one further conjecture connecting the signature axis to the volumetric-barycentric duality identified in [5].

- P1.** *Eisenstein positivity.* Verify the analogue of the Hoffreumon-Woods main theorem for the Eisenstein twisted tensor product constructed in Section 4. Show that the cone of $\mathbb{Z}[\omega]$ -Hermitian (or cyclic-invariant) operators is preserved under the twisted tensor product and that the resulting real-number quantum theory reproduces the statistics of standard complex quantum theory restricted to Eisenstein subrings.
- P2.** *Krein-space split QT.* Develop the split-complex and split-Eisenstein twisted tensor products as real-number Krein-space quantum theories. Address Question 5.1: what is the appropriate notion of state in indefinite signature, and does the twisted tensor product preserve the corresponding Krein-positivity cone?
- P3.** *Signature tension in chiral gravity.* Develop the Lorentzian Hurwitz programme based on $\mathrm{M}(2, \mathbb{Z})$, or $\mathrm{SL}(2, \mathbb{Z})$, as the maximal-order analogue, and determine whether it can carry both the volumetric trace-free condition of the $\mathfrak{sl}(2, \mathbb{C})$ sector and the barycentric trace-free condition that the asymmetric Hurwitz extension proposes.
- P4.** *The modular bridge.* Address Question 6.1: is there a real-number quantum theory based on $\mathrm{SL}(2, \mathbb{Z})$ -twisted tensor products that contains the Hoffreumon-Woods construction at $\tau = i$ and the Eisenstein companion at $\tau = \omega$, with the two compact arithmetic structures appearing as elliptic-point stabilisers inside a non-compact modular ambient?
- P5.** *Volume, balance, and signature.* The companion paper [5] identified the volumetric-barycentric duality as the structural axis distinguishing the $\mathfrak{sl}(2, \mathbb{C})$ -style trace-free condition from the $\mathbb{Z}[\omega]$ -style one. The present paper identifies the signature axis as the structural axis distinguishing definite from indefinite norm forms. The two axes are partially independent: each of the four combinations (volumetric-Euclidean, barycentric-Euclidean, volumetric-Lorentzian, barycentric-Lorentzian) is occupied by one of the

constructions of Section 4. We conjecture that the correct framework for the asymmetric extension of chiral gravity sits at the intersection of these axes, in the sense that all four constructions are simultaneously embedded as local-coordinate views of a single unified arithmetic substrate. Identifying this substrate is the deepest open problem of the present programme.

- P6.** *Methodological convergence and flag generalisation.* The Barrios Hita et al. construction [3] implements the $U(1)$ phase ambiguity of complex quantum theory via a two-dimensional flag space attached to each subsystem, together with an equivalence relation on the multipartite flag state. The split- and Eisenstein-cyclotomic constructions of Section 4 would correspond, on the flag side, to flag spaces of different dimensional and group-theoretic character: a $\mathbb{Z}_3$ -cyclic flag for Eisenstein, a non-compact $SO(1,1)$ -boost flag for split-complex, and a D_∞ flag for the golden-ratio case, with the equivalence relations on the multipartite flag states modified accordingly. Whether these flag-based constructions reproduce, on the Barrios Hita methodological side, the same theories that the twisted tensor products of Section 4 produce on the Hoffreumon-Woods side, is a question of structural compatibility between the two methodologies. A positive answer would confirm that the cyclotomy and signature axes are methodology-independent features of real-number quantum theory.

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The technical engine of this paper are constructions of Hoffreumon and Woods [1] and Barrios Hita and collaborators [3]. Both papers prompted addendums to the asymmetric Hurwitz extension [5]. The overlap between their constructions and the volumetric-barycentric duality identified in the companion paper [4] is what suggested the present paper.

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